Global multi-crop on-farm trial data from citizen science with the tricot approach

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Abstract

We present the largest open dataset of decentralised on-farm agricultural experiments, produced with the tricot approach (triadic comparisons of technology options) by 27 research organisations across 20 countries in Africa, Asia and Central America. The release holds 669 trial files covering 42 crops, 100,787 farmer-managed sites and 3.28 million plot-level observations, of which 2.44 million are rankings and 0.51 million are measurements, collected between 2012 and 2026. Each trial is published as one self-describing JSON file that carries its design, its sites with displaced coordinates, its observations, its genotypes and its full provenance, with producer, contributors and funder identified through persistent identifiers. Trait names are harmonised across trials and mapped to the Crop Ontology where terms exist. Researcher-managed companion data from the same genotypes, 130 on-station and mother trials, are linked to the tricot trials they complement. The data are released under CC0 1.0 on GitHub and archived on Zenodo under a permanent concept DOI. The dataset supports research on farmer preferences, genotype by environment interaction, climate adaptation, variety release and seed systems, and it is built so that people and AI agents fetch, filter and reuse it without accounts or tooling.

Keywords. citizen science, on-farm testing, tricot approach, participatory variety evaluation, open data, FAIR data, crop improvement, seed systems

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Background and summary

Agricultural innovation depends on evidence about how technologies perform across heterogeneous environments and under the management of their intended users[1,2]. Generating that evidence at scale and at fine spatial resolution has been difficult with conventional trials and with intensively supervised participatory approaches. The tricot approach was proposed in 2011 to meet this challenge through large-scale, decentralised citizen science in agriculture[3,4]. Options from a larger pool are allocated

to participants in connected incomplete blocks of three. In a variety trial, a farmer receives three coded seed lots, grows them side by side under local management and reports which performed best and worst for each trait. Because the triads overlap, the partial rankings of many farmers form one connected comparison network, and the relative performance of every option can be estimated although nobody evaluates the full set[3,5]. Bradley-Terry models served the first analyses and Plackett-Luce models the later ones[3,6,7].

The first pilots in India, East Africa and Central America tested barley, durum wheat, sorghum and common bean[3]. They led to the practical refinements still in use, among them package labels A, B and C instead of numbers, separate best and worst questions, pictorial observation cards and sequentially balanced allocation that tolerates attrition during seed distribution[3]. The approach then proved that distributed comparative observations support inference across environments. Rankings from 12,409 farmer-managed plots of common bean in Nicaragua, durum wheat in Ethiopia and bread wheat in India, linked to agroclimatic and soil covariates, revealed crop-specific effects of seasonal temperature on varietal performance and supported climate-informed recommendations[5]. In Ethiopia, rankings from 1,165 farmer fields predicted across-season performance of 41 durum wheat genotypes more accurately than a genomic prediction benchmark trained on station data[1]. In Ghana and Uganda, 1,433 participants ranked boiled sweetpotato in central and home tastings, extending tricot to consumer preferences and product quality[8]. Agreement among 35 Honduran farmers who ranked the same bean plots varied by trait, yet their aggregated rankings matched expert assessments, which shows how individually imperfect observations yield informative estimates once combined[9,10]. Later studies used the same data to map variety performance across Central America[11], to advance cassava clones in Uganda[12], to segment farmer demand for groundnut in Tanzania[13], to test product concepts before any variety exists[14] to link consumer preferences for cassava food products with laboratory measurements[15,16] and to read farmer preferences for potato in Nigeria[17].

Tricot grew in three overlapping phases. From 2011 to 2016 it was a proof of concept, carried by pilot trials and by early digital tools for trial design and data management. From 2017 to 2022 the ClimMob platform turned the method into a shared digital workflow for design, allocation, data collection, analysis and reporting[18], and applications spread across countries, crops and research questions[6]. Since 2022 the emphasis has been institutionalisation and interoperability, with common data structures, study-level metadata, Crop Ontology mappings, crop-specific standard operating procedures, multilingual documentation and exchange standards such as BrAPI[19,20]. The dataset described here is the data layer of that third phase.

Three features set it apart from the trial datasets that breeding programmes usually release. First, it was built from the bottom up. Every trial was designed, run and

owned by the organisation that needed the answer, a national research institute, a CGIAR centre, an NGO, a seed company or a university, with farmers as the experimenters. The dataset exists because those organisations trusted a shared method and a shared platform for a decade, and that trust is written into the files. Each trial carries its producer with a ROR identifier, its principal investigator, its contributors and its funder, so credit flows to the people who made the data wherever the data travel. Second, it is wide rather than deep. One trial of 150 farms says little on its own. Several hundred trials on 42 crops across 20 countries and 15 seasons let analysts ask which varieties perform where, how preferences differ between women and men, how performance responds to climate and whether on-farm and on-station results agree. Third, it is FAIR in full[21] and built to be used by people and by AI agents alike. One index lists every trial with the path to fetch it, one JSON file per trial describes itself completely, and nothing requires an account, a token or a database client.

The release described here is the second version of the dataset. Version 1 (2025) held 62 ClimMob v3 trials from six African countries. Version 2 adds 366 legacy trials recovered from ClimMob v2 exports, curated spreadsheets and partner files dating back to 2012, 111 further ClimMob v3 trials, and 130 researcher-managed companion trials on the same genotypes. It moves the whole collection to one schema, one naming rule and one licence. It is also the first release that reaches beyond Africa, to India and Central America, where tricot began.

Methods

The tricot design

The unit of the design is the set of three technology options evaluated by one participant. The options are most often crop varieties and breeding lines, and sometimes food products, agronomic practices or product concepts. The pool usually holds 8 to 12 comparable options and never fewer than two; in this release the median trial compares 10 genotypes and the largest 48. Each participant receives one incomplete block of three, which keeps the entry threshold low in land, time and training. Blocks are allocated at random by ClimMob in a sequentially balanced way, so that each option appears at roughly equal frequency and every village receives a balanced set even when distribution stops early[3,18]. Packages are coded A, B and C and the identity of the options is revealed only after the data are collected, to prevent bias toward familiar names.

Participants grow the three options at the same time in adjacent plots of equal size, under their own management, which maximises external validity. At defined crop stages they record on pictorial observation cards which option is best and which is worst for each trait, for instance germination, vigour, pest and disease resistance, drought tolerance, yield, taste and marketability, and for overall preference. Many

trials also compare each option with the farmer's own local variety, and a growing share records measured traits such as plot yield or days to maturity. Field agents, usually extension officers, digitise the cards with the ODK Collect app and submit them to ClimMob, where the project owner keeps full rights over the data[18]. Ranks are stored as the position of each option within the site, 1 being best; the middle rank is inferred from the best and worst answers. Rankings are analysed with Plackett-Luce models, which estimate for each option the probability of outranking the others[7,22], and with model-based recursive partitioning to find the climatic, geographic or socio-economic covariates that change the ranking[5,23].

Sources of the data

The release combines four kinds of sources (Table 1). ClimMob v3 projects were exported from the platform database with their project metadata, registration survey, data collection forms and package allocation. ClimMob v2 projects, run before 2020, survive as raw exports, as curated spreadsheets prepared for earlier publications[5,11] and as JSON exports of the old platform. Partner files cover trials run with the tricot design outside ClimMob, among them the national on-farm trials of Integrated Seed Sector Development Ethiopia, the NextGen Cassava trials in Nigeria and Tanzania and the consumer and concept tests run with video-based ranking tools[14,15]. Researcher-managed companion data come from breeding databases and spreadsheets of the same programmes, 96 on-station breeding trials of cassava in Nigeria and groundnut in Tanzania and 34 participatory varietal selection mother trials of bread wheat and rice in India, which evaluate the same genotypes under researcher control.

Table 1. Sources of the trial files in release v2.

Source platform	Trial files	Sites	Years
ClimMob v2 (raw exports, curated spreadsheets, JSON exports, two trials collected on paper)	336	53,422	2012 to 2021
ClimMob v3 (database dump, plus 15 projects available only as spreadsheet exports)	190	36,479	2020 to 2026

Source platform	Trial files	Sites	Years
Partner spreadsheets and video-based ranking surveys (tricot design outside ClimMob)	13	10,886	2013 to 2025
Breeding database exports and PVS spreadsheets (researcher-managed companion data)	130	596	2011 to 2023
Total	669	101,383	2011 to 2026

From source to release file

One pipeline, kept under version control in the private working repository of the data managers, rebuilds every release file from its source in one run. Nothing in the published files is edited by hand. The pipeline has seven steps, and each step has a decision record that states the question, the method, the alternatives rejected and what the result does and does not support. The records are published with the data so that a reader, or an AI agent, can judge whether a choice fits their own question.

Identifiers. The trial identifier is the only identity of a trial. ClimMob v3 projects keep their 12-character hexadecimal project id. Legacy trials receive the first 11 hexadecimal characters of the MD5 hash of the country code and the original project code, followed by the letter v. The id has the same length and alphabet as a v3 id, cannot collide with one and reproduces itself on every run, which is how duplicates across sources were found. Sites are identified as trial id plus package number, and plots as the package letter A, B or C.

Harmonisation of traits and variables. Every question asked in a trial was mapped to a canonical question in a harmonised question library of 414 entries that the protocols team maintains for all tricot crops[24]. The published trait name is the cleaned canonical code, so a yield ranking from Nicaragua in 2014 and one from Rwanda in 2024 share one name and one identifier. The label as asked is preserved in `trait_original`, and the identifier in `trait_id`, so every harmonised name traces back to its source. Where a Crop Ontology term exists for the crop and trait, its identifier is attached to the variable definition[20]. Labels with no canonical match keep an English name and an empty identifier, and are listed as open questions for the producer.

Personal data. Names, relatives' names, phone numbers, device identifiers, photographs, enumerator identities and free text are never exported. Administrative names and local variety names are kept. Gender and age are published as categories. Coordinates are checked against the national boundary, and points outside the country, with swapped axes or a flipped sign are removed and flagged. Every site coordinate is then displaced to a random point within 1,500 m, reproducibly per site id, and redrawn if it leaves the country. Sites published in version 1 keep their version 1 coordinates so that no site receives two independent displacements. Sites without a coordinate take the mean of other located sites in the same village or district before displacement, with the source and the spread of the pool recorded in `coordinate_source` and `coordinate_uncertainty_m`. In this release 86,180 sites carry a point and 15,192 do not, the latter mostly consumer and concept tests and early India trials that recorded no village name.

Direction of rankings. Some legacy forms asked which option was worst in the first column and best in the second, and some traits, such as faster maturity, can be valued in either direction. A check compares, at every site, the option ranked best for a trait with the option ranked best overall. Traits farmers value agree with the overall choice well above chance, and the median agreement of a trait across all trials describes how it normally behaves. A trait that agrees elsewhere but, in one trial, is the overall worst on more than half of at least 40 sites is reversed and every affected row is flagged `direction_inverted`. Ten trait and trial pairs were reversed. The same check exposed three bread wheat trials from India in 2014 whose answers are shifted across columns and cannot be repaired without the paper forms; they are published with a `consistency_warning` in their metadata.

Producers, contributors and funders. Where the source recorded no producer, the principal investigator of the ClimMob project and the project code decide, with the confidence and the evidence recorded in a reviewed table. An inferred producer carries an `attribution` note in the file, so a reader tells it from a stated one. Organisations were resolved against the Research Organization Registry; 20 of the 27 producers have a ROR identifier. Contributors follow the DataCite contributor types, with the authors of the publication that first used the data as project members, the person who curated the data as data curator and the data managers as such. Funders carry the award title and number where known; 204 trials name their funder, the largest being the CGIAR Research Program on Climate Change, Agriculture and Food Security (116 trials), the Gates Foundation (49), the Global Environment Facility (22), Norad (21), BMZ (20), SDC (20) and USAID (18).

Companion data. Researcher-managed trials on the same genotypes are published in the same schema with `pvs` or `station` in the file name and `management = researcher-managed` in the index. Each companion file lists in `metadata.companion_of` the tricot trials it relates to, and each tricot file lists in `metadata.companions` the

companion files that carry data on its genotypes, with the shared genotype names. Measurements made on genotypes outside the trial design, such as laboratory assays of a food product, go to a `genotype_data` table in the tricot file itself. The three tables share the same observation columns, so an analyst joins them on `genotype_name` to fit models with item covariates.

Review before release. The processed files were reviewed as a whole before this release. Variable names were cleaned of question identifiers, genotype names of crop prefixes, spellings merged within trials, taxa given one spelling per crop and placeholders set to null. Every open question that the data could answer was answered and marked as decided, 176 proposed answers were sent to the producers for confirmation and 104 questions remain open, mostly release years of varieties, author lists, plot sizes and units. Forty further trials with 8,571 sites were processed but held back pending the approval of their producers. They will enter the next version.

Standard operating procedures

Most trials in this release predate the first published multi-crop collection of tricot standard operating procedures, pilot release v0.1 of 2026[24], which holds complete protocols for 20 crops and the core question library, and the crop-specific sweetpotato SOP published in English, French, Portuguese and Spanish[25]. The harmonised question library used here is the same one those protocols are built on, so trials run under the SOPs from 2026 onwards enter the dataset without further mapping. About 80 percent of current ClimMob trials already follow an SOP, and the data structure described here is what those SOPs produce. In both directions the protocols and the data share trait definitions, identifiers and Crop Ontology terms, which is what lets on-farm rankings sit beside on-station measurements in breeding systems[19,20].

Data Records

The release is published on GitHub at AgrDataSci/tricot-data and archived on Zenodo under the concept DOI 10.5281/zenodo.17112491[26]. The concept DOI always resolves to the latest version; the version DOI of release v2, [10.5281/zenodo.XX], identifies the exact files described in this Data Descriptor. The GitHub release is tagged with the same version, and the Zenodo record holds the full data, never only a link. All files are released under CC0 1.0.

Layout. `datasets.csv` at the repository root is the finding aid, one row per trial with the trial id, crop and taxon, country, data producer, start and end dates, trial type and management, objective, numbers of participants, sites, genotypes and observation rows, the collection moments, the ranked traits, the size in megabytes, the licence and the path to fetch. Trial files sit in `data/<crop>/` and are named `<crop>-[kind-]<iso3>-<trial_id>.json`, where `kind` is `pvs` or `station` for companion

data. Crop, country and year do not identify a trial, 313 of the 366 legacy files share all three with another file, so the trial id closes every name and no user composes a file name by hand. `metadata/` holds the JSON Schema, the data dictionary and the DataCite metadata of the release; `decisions/` the decision records of the pipeline; `docs/` the map and this paper. The release is 1.2 GB of minified JSON. A single trial file is read with any JSON parser, and a user fetches the five trials they need, not the whole set.

File structure. Every trial file follows release schema v2.0 with five parts (Table 2). `metadata` describes the trial, `site_data` has one row per site, `plot_data` is the long table of observations, `genotype_data` holds measurements on genotypes outside the trial design and `rank_analysis` is reserved for Plackett-Luce estimates. Site and plot rows link through `site_id`, genotype rows through `genotype_name`, and `metadata.variables` lists and describes every column of the three tables with its harmonised question id, Crop Ontology term, value type and unit. Every file states its schema version, the version and date of the release, the export script and interpreter that produced it, the concept DOI and the licence, so a file found anywhere identifies the dataset it belongs to.

Table 2. Organisation of the trial files, release schema v2.0.

Component	Record unit	Principal contents
metadata	One object per trial	Trial id, code, title, type, objective, source platform and collection moments; start and end dates; bounding box and location privacy method; data producer with ROR id, principal investigator and programme; contributors with ORCID and DataCite type; crop and taxon; participants by gender; genotypes with role, release year and origin; funders with award; links to companion files; one definition per variable with harmonised id, ontology term, value type and unit; changelog

Component	Record unit	Principal contents
site_data	One row per site (farm, household or station plot)	site_id, trial id, country, crop, displaced longitude and latitude with source, uncertainty and flag, adm1 to adm3, planting and harvest dates, plot area, registration and survey variables
plot_data	One row per site, plot, collection moment and trait	site_id, plot (A, B, C or station plot), genotype_name, collection moment and days after planting, trait, trait_id, trait_original, value, value_type (rank, numeric, categorical, date, vs_local, multiple), unit, flag
genotype_data	One row per genotype, trait and replicate	Measurements made outside the trial design, such as laboratory assays of the product
rank_analysis	One row per genotype, trait and moment	Plackett-Luce estimates, empty until fitted by the data managers

Coverage. Release v2 holds 669 trial files from 20 countries and 42 crops, produced by 27 organisations (Table 3, Figure 1). The 539 farmer-managed tricot trials comprise 527 on-farm variety trials, 8 consumer tests and 4 concept tests, with 100,787 sites, each the farm or household of one participant, and 3.26 million observation rows. The 130 researcher-managed companion trials add 596 station and mother-trial sites. Across all files `plot_data` holds 3,284,321 rows, of which 2,439,450 are rankings, 509,005 numeric measurements, 99,465 categorical answers, 93,403 dates and 90,511 comparisons with the local variety, and `genotype_data` holds 7,200 rows. Participants were 30,502 women, 40,423 men and 30,362 whose gender was not recorded. Participant identifiers are specific to each trial, so counts summed across trials are not counts of unique people. The trials cover 7,511 genotype entries, 451 published trait names and 414 harmonised question identifiers, 40 of them mapped to Crop Ontology terms. Ethiopia (227 trials, 24,887 sites), India (67 trials, 21,804 sites) and Tanzania (48 trials, 15,265 sites) hold the most sites, and common bean, bread wheat,

maize, groundnut and rice the most among crops. Trials started every year from 2012 to 2026. The dataset grew in two waves, the pilots and national programmes of 2013 to 2014 and the ClimMob v2 expansion of 2017 to 2019, and has added a steady 7,500 to 10,000 sites per year since 2021 (Figure 2).

Table 3. Crops, countries and sites of the farmer-managed tricot trials in release v2. Years are the start years of the trials. Companion trials are listed in the index with `management = researcher-managed`.

Crop	Taxon	Countries	Trials	Sites	Years
Bread wheat	*Triticum aestivum* L.	ETH, IND, RWA	54	17,574	2012 to 2024
Common bean	*Phaseolus vulgaris* L.	CRI, ETH, GTM, HND, IND, MOZ, NIC, RWA, TZA, UGA	98	11,024	2012 to 2024
Maize	*Zea mays* L.	ETH, GHA, GTM, HND, KEN, RWA, TZA, UGA, ZMB, ZWE	55	10,507	2015 to 2025
Groundnut	*Arachis hypogaea* L.	GHA, NGA, TZA, UGA, ZMB	22	10,174	2016 to 2025
Rice	*Oryza sativa* L.	IND	26	8,126	2013 to 2019
Sorghum	*Sorghum bicolor* (L.) Moench	ETH, KEN, NGA, RWA, TZA	37	6,280	2013 to 2025
Sweetpotato	*Ipomoea batatas* (L.) Lam.	GHA, MOZ, UGA	35	4,901	2020 to 2026
Teff	*Eragrostis tef* (Zucc.) Trotter	ETH	44	4,868	2017 to 2019
Amaranth	*Amaranthus* spp.	BEN, IND, MLI, TZA	13	4,088	2019 to 2024
Cassava	*Manihot esculenta* Crantz	CMR, COD, NGA, RWA, TZA, UGA	13	3,246	2019 to 2024

Crop	Taxon	Countries	Trials	Sites	Years
Cowpea	*Vigna unguiculata* (L.) Walp.	BFA, GHA, NGA	12	2,716	2021 to 2025
Finger millet	*Eleusine coracana* (L.) Gaertn.	ETH, IND, KEN, UGA	19	2,476	2017 to 2019
Faba bean	*Vicia faba* L.	ETH	19	2,129	2017 to 2019
Durum wheat	*Triticum turgidum* subsp. *durum* (Desf.) Husn.	ETH	13	2,113	2013 to 2019
Barley	*Hordeum vulgare* L.	ETH, IND	19	1,958	2017 to 2019
Okra	*Abelmoschus* spp.	BEN, MLI	4	1,874	2022 to 2023
Jute mallow	*Corchorus* spp.	BEN, MLI	4	1,292	2022 to 2023
Forages	mixed species	ETH, RWA	3	1,070	2022 to 2023
Potato	*Solanum tuberosum* L.	NGA, RWA	7	950	2020 to 2023
Pearl millet	*Cenchrus americanus* (L.) Morrone	IND, NGA	5	680	2019 to 2025
Chickpea	*Cicer arietinum* L.	ETH, TZA	8	564	2017 to 2024
Mixed vegetables	several	KEN	1	370	2018
African eggplant	*Solanum aethiopicum* L.	TZA	1	300	2024
Soybean	*Glycine max* (L.) Merr.	ETH, RWA, UGA	3	263	2019 to 2024
Chili pepper	*Capsicum* spp.	BEN	1	200	2024

Crop	Taxon	Countries	Trials	Sites	Years
Tomato	*Solanum lycopersicum* L.	BEN	1	200	2024
Field pea	*Pisum sativum* L.	ETH	3	171	2018 to 2019
Lentil	*Lens culinaris* Medik.	ETH	1	94	2018
Sesame	*Sesamum indicum* L.	ETH, IND	2	89	2019
Black cumin	*Nigella sativa* L.	ETH	2	73	2018 to 2019
Tepary bean	*Phaseolus acutifolius* A.Gray	GTM	1	63	2017
Noug	*Guizotia abyssinica* (L.f.) Cass.	ETH	1	50	2018
Kodo millet	*Paspalum scrobiculatum* L.	IND	1	47	[XX]
Little millet	*Panicum sumatrense* Roth ex Roem. & Schult.	IND	1	46	[XX]
Pigeonpea	*Cajanus cajan* (L.) Millsp.	KEN	1	41	2024
Moth bean	*Vigna aconitifolia* (Jacq.) Marechal	IND	1	40	2019
Mung bean	*Vigna radiata* (L.) R.Wilczek	IND	1	40	2019
Oat	*Avena sativa* L.	ETH	1	33	2019

Crop	Taxon	Countries	Trials	Sites	Years
Black gram	*Vigna mungo* (L.) Hepper	IND	2	20	2019
Small millets	*Paspalum scrobiculatum* L.; *Panicum sumatrense*	IND	2	20	2019
Rice bean	*Vigna umbellata* (Thunb.) Ohwi & H.Ohashi	IND	1	12	2019
Horse gram	*Macrotyloma uniflorum* (Lam.) Verdc.	IND	1	5	2019
Total		**20 countries**	**539**	**100,787**	**2012 to 2026**

Table 4. Summary of release v2.

Indicator	Value
Trial files	669 (539 farmer-managed tricot, 130 researcher-managed companion)
Countries	20
Crops	42
Data-producing organisations	27 (20 with ROR identifier)
Sites	101,383 (100,787 tricot, 596 companion)
Sites with coordinates	86,180
Genotype entries	7,511
Observation rows in plot_data	3,284,321 (2,439,450 ranks, 509,005 numeric)
Rows in genotype_data	7,200
Harmonised question identifiers	414, with 451 published trait names
Participants by gender	30,502 women, 40,423 men, 30,362 not stated
Trials with a named funder	204

Indicator	Value
Trials held in quarantine pending approval	40 (8,571 sites)
Size	1.2 GB, minified JSON
Licence	CC0 1.0

Technical Validation

Design integrity. Every tricot site holds exactly three plots of distinct genotypes from the trial pool, and every ranking is a complete order of those three. Ties, incomplete answers and package codes used by two participants with different answers were dropped at export and counted in the export reports, so no ranking in the release is ambiguous. Legacy trials with fewer than five ranked sites were not published (391 ClimMob v2 projects holding 30 ranked records in all), nor were 17 Ethiopian projects shown to be copies of another project, matched package by package.

Consistency of rankings. The direction check described in Methods ran on every trait of every trial. After the ten documented reversals it returns no further candidate. The three inconsistent India wheat trials are published with a warning rather than dropped, because their site, management and yield data remain sound and the warning is machine-readable.

Coordinates. Every published point lies inside its country. Points that failed the boundary check were removed and are counted per trial in the export report. Pooled coordinates carry their source and the spread of the pool, so an analyst can filter to farm-level points by `coordinate_source` when the question needs them.

Identifiers and names. No two files map to the same path, every file carries the same trial id in its name, its metadata and the index, and no published variable name ends in a digit except the schema fields. No genotype name starts with its crop name, and spellings that differ only in case or spacing were merged within trials. Every file states a producer, and the 222 inferred producers are marked as such. Every trial has a taxon from one reviewed table.

Personal data. No value in `site_data` matches a phone number or email pattern, and no published column refers to names, phones, device identifiers or photographs. Two columns in Ethiopian trials that held personal names instead of place names were withheld.

Schema validation. Every file is validated against the JSON Schema of release schema v2.0 before the release is tagged [XX, confirm once the v2.0 schema is written and the validation run]. The index is rebuilt from the files, so its counts agree with the files by construction.

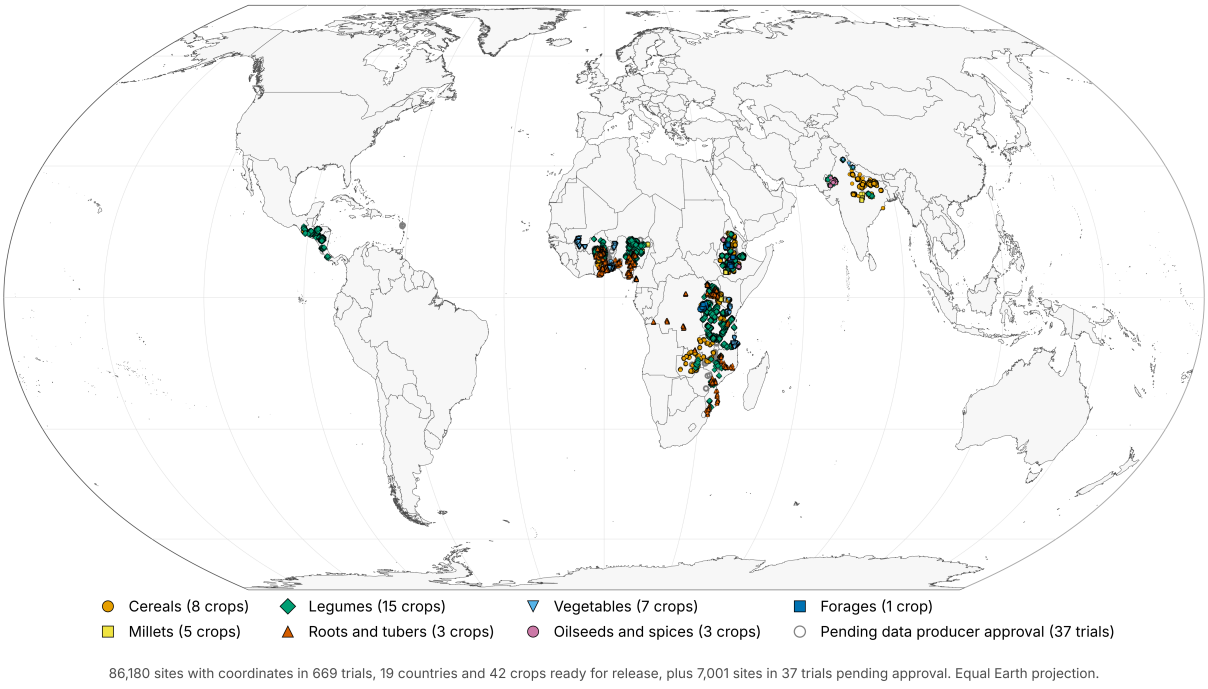


Figure 1: Figure 1. Trial sites in release v2, by crop group. Each point is one site after displacement within 1,500 m. Hollow grey rings are sites of trials processed but pending the data producer's approval. Equal Earth projection, EPSG:8857. Administrative boundaries from Natural Earth. The authors remain neutral with regard to jurisdictional claims in maps. [Rebuild with the final counts before submission.]

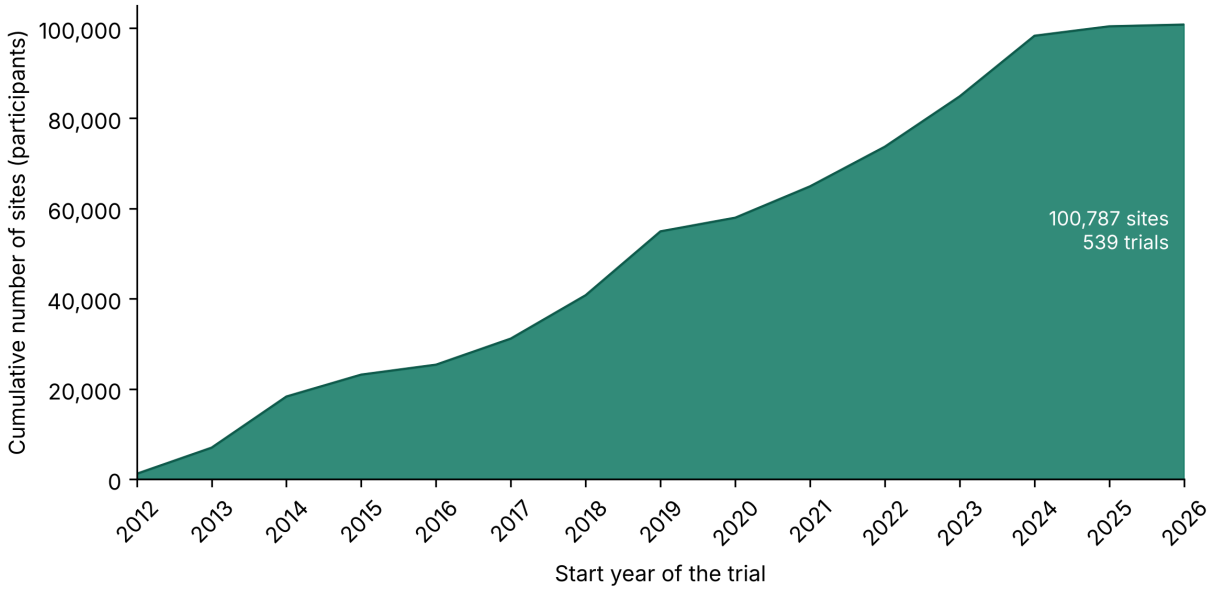


Figure 2: Figure 2. Cumulative growth of the farmer-managed tricot trials by start year. The area gives the cumulative number of sites and the line the cumulative number of trials, all 539 trials and 100,787 sites. For 44 trials without a start date in the index the year was taken from the project code or, for 18 trials, from the source project [XX, confirm the 18 estimated years in `metadata/questions/`].

Reproducibility. The whole release is rebuilt by one command from the sources, and each file records the script and interpreter version that wrote it. Independent reanalysis of the published files reproduces results of earlier studies built on subsets of the same data [XX, cite the reproduction in tricot-analysis-workflows once run, e.g. Brown et al. 2022 for the Central American beans].

Usage Notes

Finding and fetching. Start from `datasets.csv`. Filter by crop, country, year, type, management, trait or producer, then fetch the files in the `path` column from the GitHub raw URL or from the Zenodo archive. A person, a script or an AI agent needs one URL per file and no login. The repository ships a skill file (`SKILL.md`), an `AGENTS.md` and an `llms.txt` that tell an AI assistant what the data hold, how to fetch them and how to cite them. The README gives a worked example in R that reads a trial file, builds the rankings with the gosset package and fits a Plackett-Luce model with PlackettLuce[7,22]; the ClimMobTools package reads the same structure. The companion tricot-analysis-workflows repository holds the team's curated workflows, from variety recommendation and genotype by environment analysis to the integration of trial sites with climate and soil data.

Pooling trials. The harmonised trait names and identifiers make pooling straightforward, but the analyst decides what to pool. Trait wording, collection moments and local varieties differ among trials, and `trait_original`, `collection_moment` and the variable descriptions carry what is needed to judge comparability. Trials with a `consistency_warning` should be read before use in trait by trait analysis. Rankings flagged `direction_inverted` were corrected and the flag is informational.

Covariates. Displaced coordinates are fit for linking sites to gridded climate, soil and elevation data at kilometre resolution[5,11,27], and the climatrends and chirps R packages were written for that purpose. They are not farm locations. Planting dates let an analyst compute season-specific covariates, and the companion files let item covariates measured on station enter the model.

Credit and citation. CC0 places no legal obligation on users. The community norm the dataset asks for is the one GBIF asks of its users. Cite the version DOI you used and name the concept DOI so others find the current one. Cite, alongside the dataset, the publications that first used the trials you reused, which are listed in each file's metadata, and list the trial ids you used in your paper or supplement. Author lists grow with versions, since each Zenodo version carries its own creators, so producers added in a later batch appear from that version on.

Governance. Producers approve the release of their trials; 40 processed trials wait in quarantine for that approval. Personally identifiable information never leaves the ClimMob project owner's control. Local variety names are published because they are part of the agricultural knowledge the trials record, and the dataset makes no claim on the traditional knowledge they carry. The dataset is maintained by two data managers in the margins of other work, which is why the design puts everything into self-describing files and one index, with no database or API to keep alive.

What comes next. Version 3 will add the quarantined trials, the Plackett-Luce estimates in `rank_analysis`, fuller Crop Ontology coverage and the trials run under the published SOPs. Trials collected through speech-based interfaces and trials that link to station data through breeding information systems will enter the same files, since the schema already carries companion links and measured traits. The dataset is one layer of a documented ecosystem that also publishes the protocols that produce the data, the evidence built on them and the analysis workflows that use them, each with its own identifier, so that a user can move from a question to the trials behind the answer.

Code availability

The data are archived on Zenodo under the concept DOI 10.5281/zenodo.17112491 and maintained on GitHub at https://github.com/AgrDataSci/tricot-data under

CC0 1.0. The export pipeline and its decision records are in the same repository under `script/` and `decisions/` [XX, confirm what is copied from the private working repository]. The R packages gosset, ClimMobTools, climatrends and chirps are on CRAN under the MIT licence. ClimMob is open source at https://github.com/BioversityCostaRica [XX, confirm URL].

Author contributions

[CRediT taxonomy, to complete with the final author list.] Conceptualization, K.d.S., J.v.E.; data curation, K.d.S., M.-A.L.; methodology, K.d.S., M.-A.L., A.C.; software, K.d.S., B.Y.; investigation and resources, all data producers; writing, original draft, R.C., A.C., K.d.S.; writing, review and editing, all authors; supervision and funding acquisition, J.v.E., K.d.S.

Competing interests

The authors declare no competing interests.

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